barrier excludes most macromolecules, microorganisms, immunocompetent cells and antibodies. Even pathogens that are circulating in the bloodstream usually cannot enter the brain and spinal cord because of blood brain barrier. Certain capillaries permit some substances to pass from the blood into the brain but restricts others. These capillaries are less permeable than others within the body and are therefore more selective in passing materials (Figure 12.20). The blood brain barrier (Figure 12.21) is due to the cellular configuration of cerebral capillaries, the choroid plexus and arachnoid cells. It acts as a natural barrier that prevents the invasion of microorganisms into the brain. If this is breached organisms enter the brain. The blood CSF barrier (Figure 12.22) (also calle brain CSF barrier) consists of endothelium with fenestrations, and tightly joined choroid plexus epithelial cells. It acts as a natural barrier that prevents the invasion of microorganisms into the meninges.

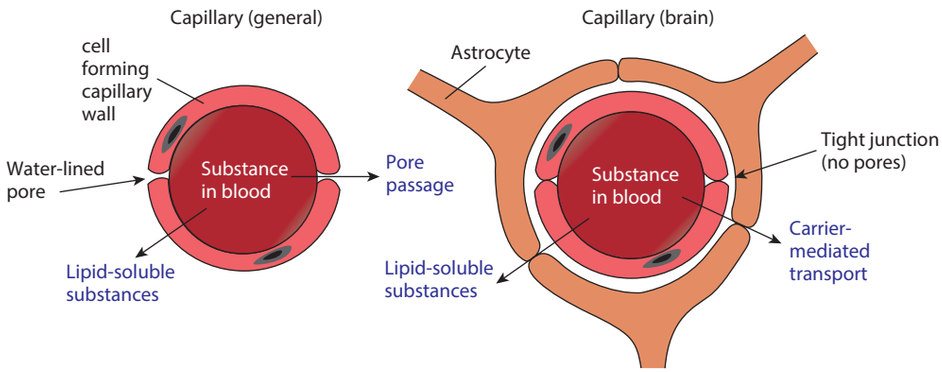


Figure 12.20: Capillaries of brain

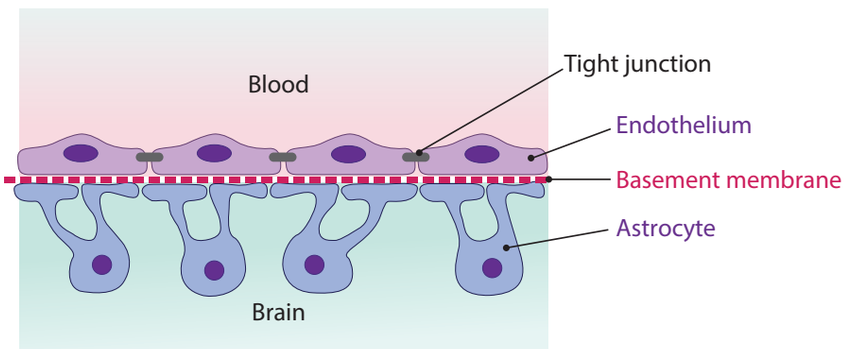


Figure 12.21: Blood brain barrier

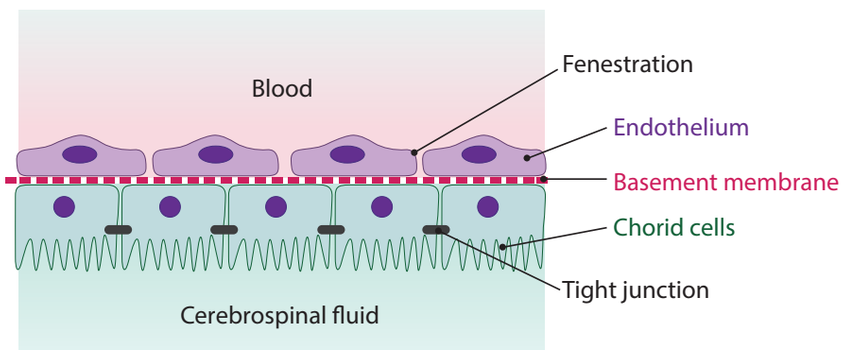


Figure 12.22: Blood CSF barrier

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